

Effective-Group Orbits of Finite Repair Profiles in a Frozen Hénon Core

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Abstract

The ambient symmetry lift of the frozen named core has order 11520, but its label action has a six-element kernel. We use the faithful 1920-element image to quotient all 65536 deletion masks while retaining a finite repair profile: C79's distance and witness multiplicity, generated-subgroup order, and C80 threshold histograms grouped by target subgroup order. The quotient has exactly 3024 orbits and fourteen profile classes. A weighted orbit polynomial recovers $(1+x)^{16}$, and a fixed-support identity independently recovers the orbit count. No complete Burnside ring, table of marks, arithmetic, or Hilbert–Polya claim is made.

1 Two group orders

C75 records an ambient lift of order 11520. Its action on the sixteen named labels has a kernel of order six, so the effective permutation group used here has order

$$|E| = 11520/6 = 1920.$$

This distinction is substantive: orbit sizes and stabilizers below are those of E , not of the nonfaithful lift.

2 Invariant repair profile

For a retained mask A and deletion set $D = L \setminus A$, let

$$\Pi(D) = (\rho(D), W(D), |\Phi(A)|, H_D),$$

where H_D lists, for each of the eight distinct subgroup orders, the counts of C80 thresholds 0, 1, 2, 3 among the target rows of that order. An effective label automorphism preserves $|\Phi(A)|$ and permutes targets within order classes, while C79's (ρ, W) is preserved. Thus Π is constant on E -orbits.

Proposition 1. *The effective action partitions the 65536 masks into 3024 orbits. Their size spectrum is*

$ O $	1	2	4	5	8	10	16	20	40	80	160
$\#O$	128	256	416	128	192	384	16	672	608	208	16

The weighted orbit inventory is exact: if $r(O)$ is a representative's cardinality, then

$$\sum_O |O| x^{r(O)} = (1+x)^{16}.$$

The profile quotient contains fourteen distinct profile classes. In particular, the ‘(rho,W)’ mask marginal remains

$$(0, 1) : 30400, (1, 1) : 30400, (1, 4) : 1984, (1, 7) : 192, (1, 8) : 128, \\ (2, 4) : 1984, (2, 7) : 192, (2, 8) : 128, (2, 25) : 64, (3, 25) : 64.$$

3 Orbit-count audit

For a permutation $g \in E$, the number of fixed label subsets is $2^{c(g)}$, where $c(g)$ is its cycle count on sixteen labels. The stored cycle-count spectrum therefore gives the elementary identity

$$\frac{1}{1920} \sum_{g \in E} 2^{c(g)} = 3024.$$

This is an orbit partition check only; it is not a computation of a full Burnside ring or table of marks.

4 Certification and scope

The producer and checker bind C76/C78/C79/C80 evidence and manifests by raw bytes, rebuild the five generators and closure, and compare every orbit representative, size, stabilizer, and profile. The canonical evidence hash is `c3cc35f45e1c8f7c9d4ecaecca820bf9dbc4db1c6a5769c20c75bad21f32fd9f`. A separate SymPy check verifies the weighted polynomial, class partition, and fixed-support identity; clean replay preserves the digest and all fourteen hostile mutations are rejected. The result is a finite named-coordinate quotient under the explicit scope firewall `‘NO_BAD_EULER_OR_ROOT_NUMBER’`.